IV Bolus Practice 2

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

a 250 mg of an antibiotic was administered as an IV bolus injection to a 70-kg patient. The concentration at 4 and 8 hours after administration of the dose was 6.6 mcg/mL and 2.18 mcg/mL, respectively. What is the elimination rate constant?

b An antibiotic given as a 250 mg IV bolus injection has an elimination rate constant of 0.2769 hr^{-1}. What is its half-life of elimination?

c An antibiotic given as a 250 mg IV bolus injection has an elimination rate constant of 0.2769 hr^{-1} and a plasma concentration of 2.18 mcg/mL at 8 hours. What is the initial plasma concentration?

d An antibiotic given as a 250 mg IV bolus injection has an initial plasma concentration of 20 mg/L and an elimination rate constant of 0.2769 hr^{-1}. What is the concentration of drug in the plasma 15 minutes after the dose was given?

e A 250 mg intravenous bolus dose of an antibiotic produces an initial plasma concentration of 20 mg/L in a 70-kg patient. What is the apparent volume of distribution?

f An antibiotic has an apparent volume of distribution of 12.5 L and an elimination rate constant of 0.2769 hr^{-1} in a patient. What is the total body clearance of this drug in this patient?

g An antibiotic was given as a 250 mg IV bolus injection and has an elimination rate constant of 0.2769 hr^{-1}. What is the amount of drug in the body 3 hours after the drug was administered?

h An antibiotic given by intravenous bolus injection has an elimination half-life of 2.5 hours. How much time is required for 99.9% of the drug to be eliminated from the body?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 0.2769 hr^{-1}

WHAT THE STEM GIVES

D_0	250 mg	*not needed: k comes from two concentrations alone*
weight	70-kg	*not needed*
t_1	4	the earlier time, in hr
t_2	8	the later time; $t_2 - t_1 = 4$ hr
C_{p^1}	6.6 mcg/mL	the numerator of the ratio, as 6.6 mg/L
C_{p^2}	2.18 mcg/mL	the denominator of the ratio, as 2.18 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(Cp1/Cp2)}{t_2 - t_1}$$

Two concentrations at two times define the slope of the log-linear decline, and the rate constant is that slope with the sign reversed. The earlier and larger concentration goes on top so the logarithm is positive, which is what keeps k from coming out negative.

UNIT

6.6 mcg/mL = 6.6 mg/L and 2.18 mcg/mL = 2.18 mg/L

Dr. Mosley tells the class to take it on her word that micrograms per millilitre equals milligrams per litre, so no arithmetic conversion is needed. Here the units cancel inside the ratio anyway, but the equivalence matters for the later parts that pair a concentration with a dose in milligrams.

ALGEBRA

$$k = \frac{\ln(6.6/2.18)}{(8 - 4)\ \text{hr}} = \frac{1.10774}{4\ \text{hr}}$$

The ratio $\frac{6.6}{2.18}$ is 3.0275 and its natural logarithm is 1.10774. Dividing by the four-hour interval between the samples converts a dimensionless logarithm into a quantity in reciprocal hours.

ROUND

$k = 0.2769\ hr^{-1}$

The quotient is 0.276936, which her key prints as 0.2769 hr^{-1}. She asks for three to four decimal places on k so that early truncation does not shift every downstream answer.

> **DOES THE NUMBER MAKE SENSE?**
> From 4 to 8 hr the concentration falls 6.6 → 2.18 mcg/mL, to about a third: more than halved but not quartered in 4 hr, so t½ is between 2 and 4 hr and k between 0.693 over 4 and 0.693 over 2; 0.2769 fits.

b

Her key

Answer: 2.5 hr

WHAT THE STEM GIVES

D_0	250 mg	*not needed: the half-life comes from k alone*
k	0.2769 hr^{-1}	the denominator of $\frac{0.693}{k}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{k}$$

Half-life is the time for the concentration to fall by half, and for a first-order process that time is fixed by the rate constant alone. The 0.693 is the natural logarithm of 2, which is what falls out of solving $C_t = C_0e^{-kt}$ for C_t equal to half of C_0.

ALGEBRA

$$t½ = \frac{0.693}{0.2769\ hr^{-1}} = 2.5027\ hr$$

Dividing a dimensionless number by reciprocal hours leaves hours, which is what a half-life must carry. The dose does not enter: a first-order half-life is the same whether 250 mg or 25 mg was given.

ROUND

$$t½ = 2.5\ hr$$

Her key prints 2.5 hr, to one decimal place. The later parts of this problem use 2.5 hr where a half-life is needed, so rounding here and rounding at the end give the same answers.

> **DOES THE NUMBER MAKE SENSE?**
> 0.693 over a quarter would be a little under 3 hr, and 0.2769 is a little more than a quarter, so t½ is about 2.5 hr.

c

Her key

Answer: 20 mg/L

WHAT THE STEM GIVES

D_0	250 mg	*not needed: the decline is worked in concentration*
k	0.2769 hr^{-1}	the exponent kt = 0.2769 × 8
C_p	2.18 mcg/mL	the 8-hour concentration, as 2.18 mg/L, multiplied by e^{kt} = 9.1632
t	8 hours	the sample time in the exponent

THE WORKING, ONE LINE AT A TIME

SETUP

$C_0 = C_t \cdot e^{+kt}$, with t the time of the measured sample

No sample can be drawn at the instant of the injection, so the initial concentration has to be recovered from a later one. Running the decay backwards makes the exponent positive, which is the one sign change the step turns on.

UNIT

2.18 mcg/mL = 2.18 mg/L

A microgram per millilitre and a milligram per litre are the same number, so nothing has to be converted. The switch of label matters here because the volume of distribution in the next part pairs this concentration with a dose in milligrams, and litres is what makes that division come out in litres.

ALGEBRA

$C_0 = (2.18\ \text{mg/L})e^{(0.2769\ \text{hr}^{-1})(8\ \text{hr})} = (2.18\ \text{mg/L})(9.1632) = 19.98\ \text{mg/L}$

Eight hours is about 3.2 half-lives, so the concentration at 8 hours is roughly a ninth of the starting value, and multiplying back by that factor recovers it. Reciprocal hours times hours leaves the exponent dimensionless.

ROUND

$C_0 = 20$ mg/L

Her key prints 20 mg/L. She writes the working with the mcg/mL figure and the answer in mg/L without comment, because the two units carry the same number.

> **DOES THE NUMBER MAKE SENSE?**
> With k = 0.2769 hr^{-1} the half-life is between 2 and 3 hr, so 8 hours is a little over three half-lives and C_0 is a little over eight times 2.18 mg/L; 20 mg/L fits.

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d Her key

Answer: 18.66 mg/L

WHAT THE STEM GIVES

D_0	250 mg	*not needed: C_0 is already given*
C_0	20 mg/L	the starting concentration, multiplied by e^{-kt}
k	0.2769 hr^{-1}	the exponent kt = 0.2769 × 0.25
t	15 minutes	becomes 0.25 hr so the exponent is a pure number

THE WORKING, ONE LINE AT A TIME

SETUP

$C_t = C_0 \cdot e^{-kt}$

The concentration at any time after a bolus is the intercept decayed for the elapsed time, and the whole of the dose is in the body at time zero. Nothing else is needed once C_0 and k are in hand.

UNIT

$15 \text{ minutes} = 15/60 \text{ hr} = 0.25 \text{ hr}$

The rate constant is in reciprocal hours, so the time has to be in hours for the exponent to come out dimensionless. She gives the interval in minutes every time this part appears and converts it silently, which is where the mark is usually lost.

ALGEBRA

$$C_t = (20 \text{ mg/L})e^{-(0.2769 \text{ hr}^{-1})(0.25 \text{ hr})} = (20 \text{ mg/L})(0.93312) = 18.66 \text{ mg/L}$$

A quarter of an hour is a tenth of a half-life, so barely any drug has left and the concentration is still 93 per cent of the intercept. That closeness to C_0 is the check that the minutes were converted rather than used raw.

ROUND

$C_t = 18.66 \text{ mg/L}$

Her key prints 18.66 mg/L, to two decimal places, matching the precision she carries on concentrations throughout this sheet.

> **DOES THE NUMBER MAKE SENSE?**
> A quarter hour is a small fraction of a half-life of over 2 hr, so only about 7 per cent is lost: C_t sits a little under 20 mg/L and well above half of it, and 18.66 mg/L fits.

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e

Her key

Answer: 12.5 L

WHAT THE STEM GIVES

D_0	250 mg	the numerator: the amount in the body at time zero
C_0	20 mg/L	the denominator
weight	70-kg	*not needed: V_D comes from the dose and the concentration*

THE WORKING, ONE LINE AT A TIME

SETUP

$$V_D = \frac{D_0}{C_0}$$

V_D is the volume that would have to hold the whole dose to give the measured plasma concentration. So it is the dose divided by the concentration at the moment the whole dose is still in the body. That moment is time zero, so the back-extrapolated intercept, not a later sample, goes in the denominator.

ALGEBRA

$$V_D = \frac{250 \text{ mg}}{20 \text{ mg/L}} = 12.5 \text{ L}$$

Milligrams divided by milligrams per litre leaves litres. Using a later concentration, such as the 6.6 mg/L at 4 hours, would give 37.9 L, because by then part of the dose has gone.

ROUND

$V_D = 12.5 \text{ L}$

Her key prints 12.5 L. The 70-kg weight is stated in the stem and never used in her solution, since the volume is asked for in litres rather than per kilogram.

DOES THE NUMBER MAKE SENSE?

250 mg over 20 mg/L is 12.5 L, roughly a fifth of the 70-kg body weight; V_D must come out in litres.

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f

Her key

Answer: 3.46 L/hr

WHAT THE STEM GIVES

V_D	12.5 L	one factor of the product $k \cdot V_D$
k	0.2769 hr^{-1}	the other factor

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl = k \cdot V_D$

Clearance is the volume of plasma cleared of drug per unit time, and it is the fraction removed per hour multiplied by the volume that fraction is taken from. Both quantities are already in hand from the earlier parts, so no new data are needed.

ALGEBRA

$Cl = (0.2769\ hr^{-1})(12.5\ L) = 3.4613\ L/hr$

Reciprocal hours multiplied by litres leaves litres per hour, which is what a clearance must carry. Each hour the equivalent of 3.46 L of plasma is stripped of drug, out of the 12.5 L the drug occupies.

ROUND

$Cl = 3.46\ L/hr$

Her key prints 3.46 L/hr, to two decimal places, which is the precision she carries on clearances throughout this sheet. As a check, dividing the clearance by the volume returns the 0.2769 hr^{-1} the set started from, so nothing has drifted through the six parts that lead here.

DOES THE NUMBER MAKE SENSE?

0.2769 is a little over a quarter, so about a quarter of 12.5 L, roughly 3 L, is cleared each hour; 3.46 L/hr fits, and k below 1 keeps Cl below 12.5 L/hr.

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g

Her key

Answer: 108.9 mg

WHAT THE STEM GIVES

D_0	250 mg	the starting amount, multiplied by e^{-kt}
k	0.2769 hr^{-1}	the exponent $kt = 0.2769 \times 3$
t	3 hours	the time in the exponent

THE WORKING, ONE LINE AT A TIME

SETUP

$Dt = D_0 \cdot e^{-kt}$

Amounts decay by the same first-order law as concentrations, with the same rate constant, because the two differ only by the fixed volume of distribution. Working in amounts here avoids computing a concentration and multiplying it back by 12.5 L.

ALGEBRA

$Dt = (250\ \text{mg})e^{-(0.2769\ \text{hr}^{-1})(3\ \text{hr})} = (250\ \text{mg})(0.43574) = 108.9\ \text{mg}$

Three hours is slightly more than one half-life of 2.5 hours, so slightly less than half the dose remains, which is what 108.9 mg out of 250 mg is. Reciprocal hours multiplied by hours leaves the exponent dimensionless.

ROUND

$Dt = 108.9\ \text{mg}$

Her key prints 108.9 mg, to one decimal place, which is the precision she carries on amounts throughout this sheet. Three hours is 1.2 half-lives, so a shade under half the 250 mg should remain, and 108.9 mg is that shade under 125 mg.

> **DOES THE NUMBER MAKE SENSE?**
> With $k = 0.2769\ \text{hr}^{-1}$ the half-life is between 2 and 3 hr, so 3 hours is a little over one half-life and a little under half of 250 mg remains; 108.9 mg fits, below half of the dose.

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h

Her key

Answer: 25 hr

WHAT THE STEM GIVES

t½	2.5 hours	multiplied by the 10 half-lives that 99.9% elimination takes
fraction	99.9%	eliminated, which leaves 0.1%, one part in 1000: 10 half-lives

THE WORKING, ONE LINE AT A TIME

SETUP

99.9 per cent eliminated leaves 0.1 per cent, and a tenth of a per cent is reached after 10 half-lives

Each half-life halves what is left, so the fraction remaining after n half-lives is one over two to the n. Ten halvings give $\frac{1}{1024}$, which is 0.098 per cent, just under the tenth of a per cent asked for.

ALGEBRA

$t = 10\ t_{½} = 10(2.5\ \text{hr}) = 25\ \text{hr}$

Multiplying the half-life by the number of halvings gives the elapsed time directly. No logarithm is needed once the count is recognised.

ALGEBRA

By the equation: $t = \frac{\ln(100/0.1)}{0.2769\ \text{hr}^{-1}} = \frac{6.9078}{0.2769\ \text{hr}^{-1}} = 24.95\ \text{hr}$

The general route divides the logarithm of the ratio of starting to remaining amount by the rate constant. It agrees with the half-life count to within the rounding, which is what confirms the shortcut.

ROUND

t = 25 hr

Her key prints 25 hr and offers both routes, the equation and the ten half-lives. The 0.05 hr difference between them is the rounding in 2.5 hr.

DOES THE NUMBER MAKE SENSE?

99.9% eliminated leaves 0.1%, one part in 1000, which is ten halvings, so t = 10 × 2.5 hr = 25 hr.